Physics AI-Based Simulation Prediction of Mechanical Components

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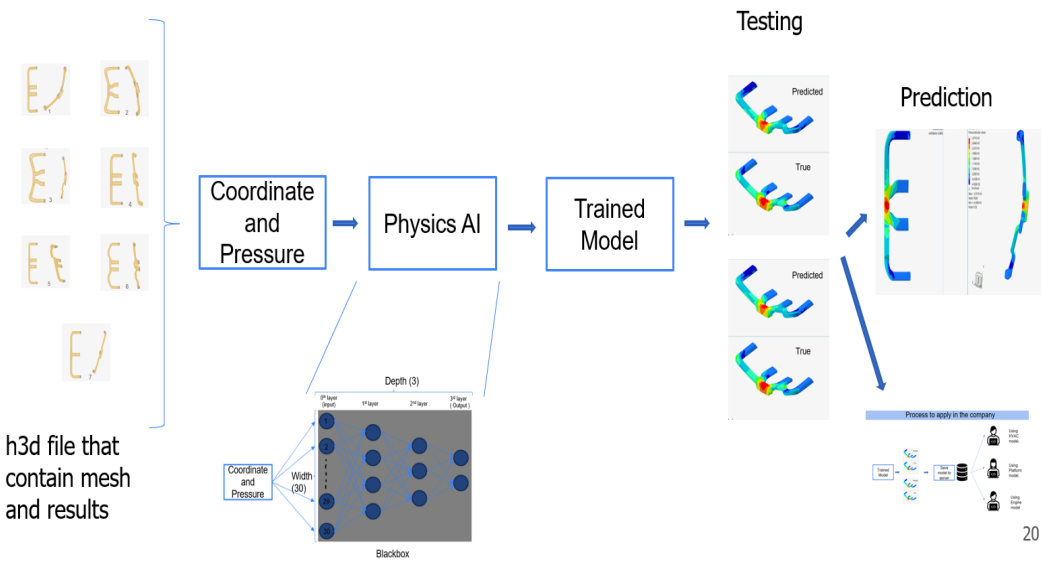
1. Objective

To develop a Physics AI based approach to predict simulation results and reduce computational time for repetitive CAE problems.

2. Workflow

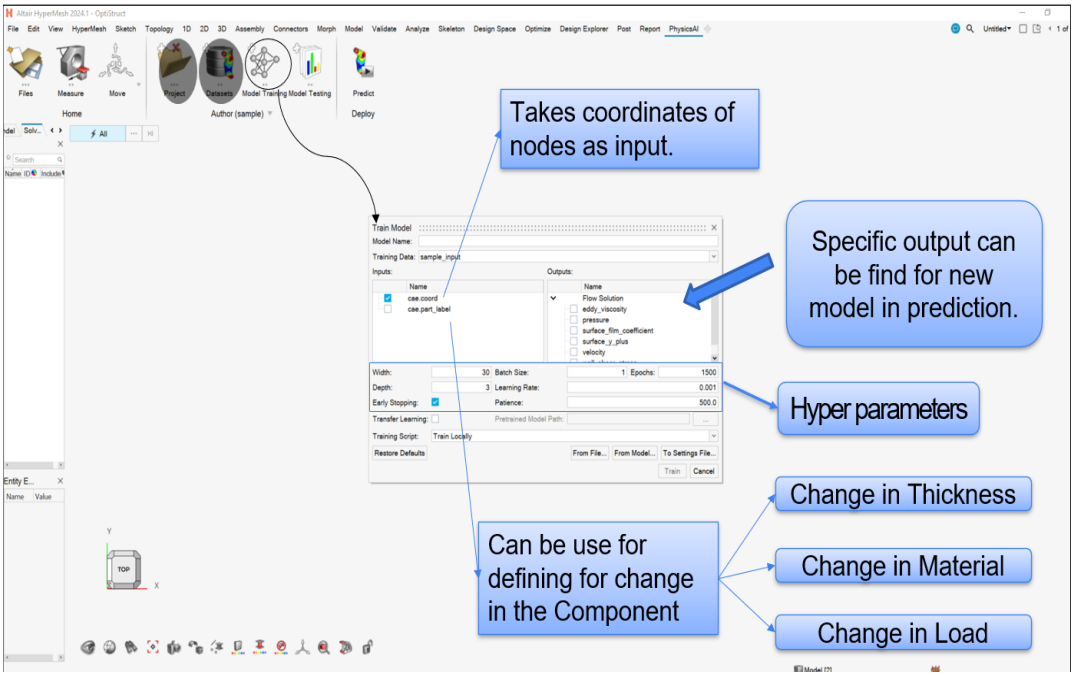
The workflow involves dataset creation, training, testing, and prediction using Physics AI.

Physics AI is not the substitute of simulation software it is complimentary to it.



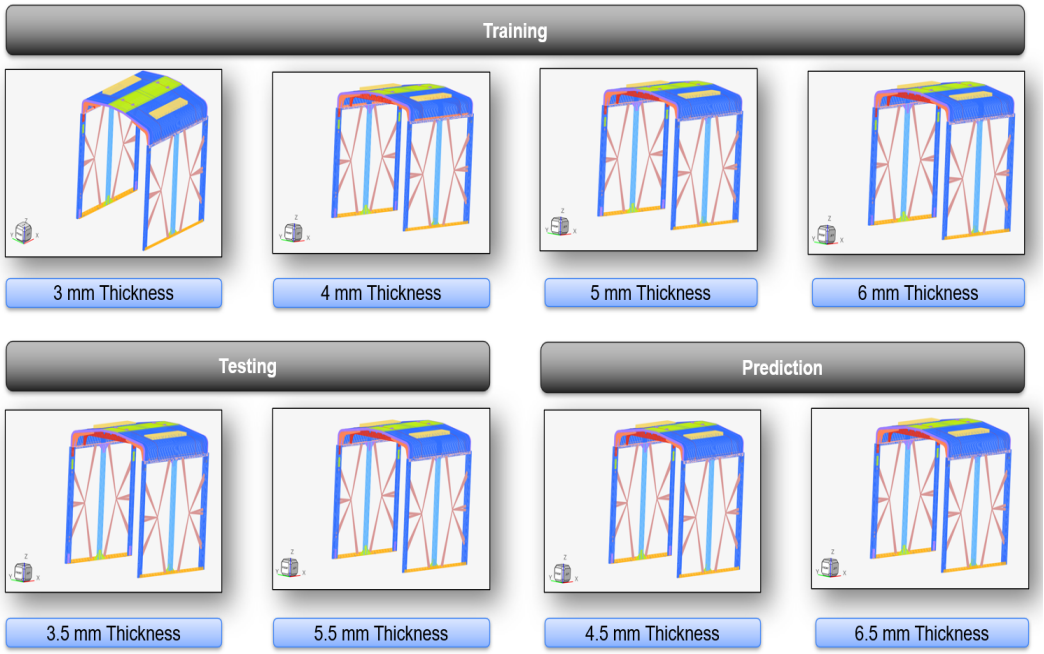
3. Model Training

Physics AI model is trained using node coordinates and simulation data with hyperparameters tuning.



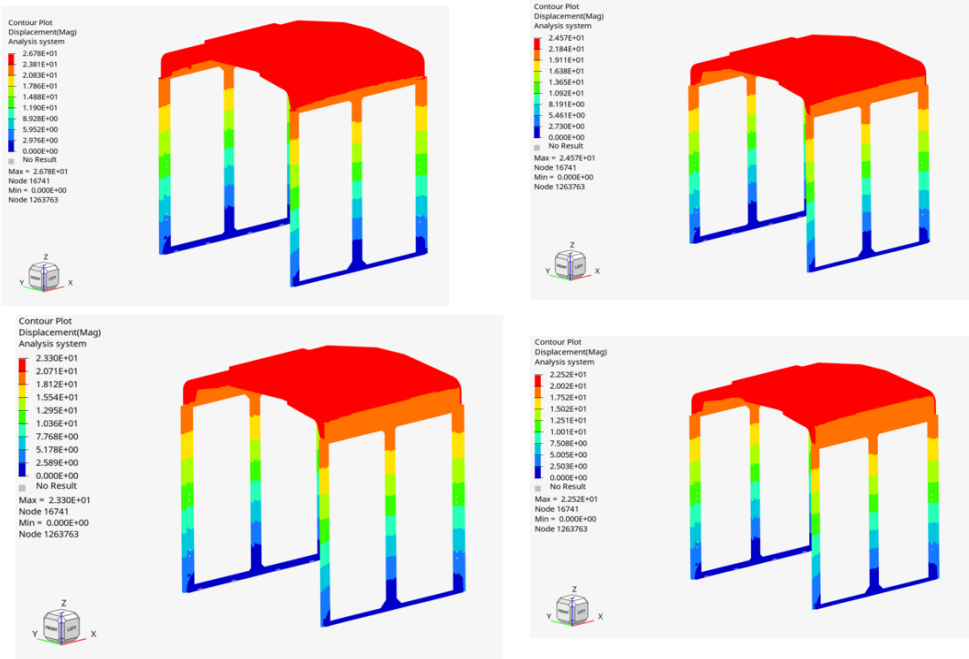
4. Dataset Variation

Different geometry and thickness variations are used for training and validation.



5. Results Comparison

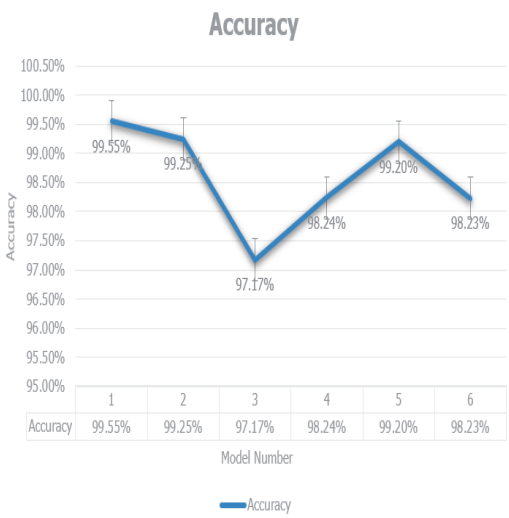
Comparison between predicted and simulated results shows high accuracy.



6. Accuracy Analysis

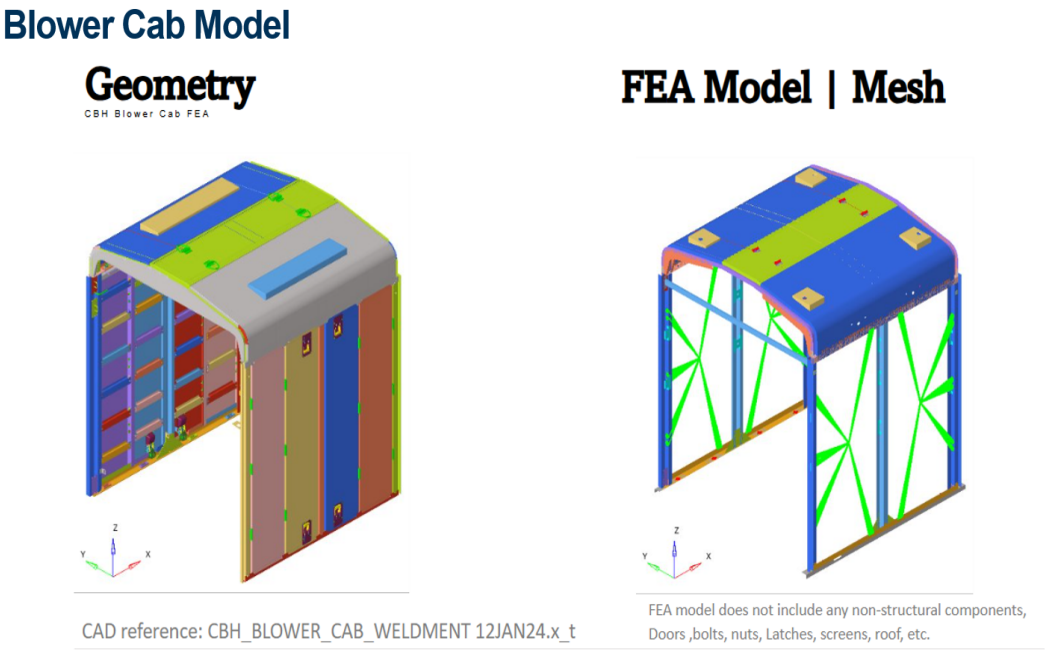
Model achieved accuracy in the range of 93%–99%.

Model No.	Accuracy
1	99.55%
2	99.25%
3	97.17%
4	98.24%
5	99.20%
6	98.23%



7. Component Model

Geometry and FEA mesh of the component used in analysis.



8. Conclusion

Physics AI significantly reduces simulation time and is effective for repetitive engineering simulations.